

Alexander K. Kariuki

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[GitHub](#) | [LinkedIn](#)

Result-driven Data Scientist with 4 years of experience delivering data-powered solutions across corporate and social impact sectors. Builds end-to-end ML and Generative AI systems - from statistical modelling and predictive analytics through to LLM engineering, Agentic AI workflows, and production deployment. Proven track record of translating complex business challenges into scalable solutions that drive measurable outcomes.

EXPERIENCE

Data Scientist

April 2024 - August 2025

Fahamu AI

- Designed and deployed RAG pipeline integrating OpenAI API with company knowledge base, enabling context-aware AI responses and reducing customer service workload by 30%
- Built predictive models using Python and Scikit-learn to forecast user behaviour patterns achieving 80% accuracy and informing product roadmap decisions
- Developed prompt templates and few-shot learning strategies for LLM applications, standardizing AI outputs across marketing and content generation workflows
- Collaborated with engineering team to implement vector search functionality using embeddings, improving document retrieval relevance by 45%

Junior Data Scientist

August 2021 - August 2023

Dalberg Data Insights

- Architected automated Python-based ETL workflow to extract and transform API data for GAVI, the Vaccine Alliance, streamlining data ingestion into PostgreSQL and reducing manual processing effort by 40%
- Developed 10+ predictive models using supervised and unsupervised learning algorithms that optimized business decision-making and increased operational efficiency by 25% across multiple client projects
- Built organization's internal React-based visualization library, standardizing data presentations across front-end applications and improving development efficiency for the 10 team members
- Co-authored technical proposals for 5+ business development initiatives (including MCC Lesotho), designing data infrastructure and analytics strategies to address complex health and development challenges
- Enhanced Tableau dashboards for KPI tracking and reporting, enabling real-time data monitoring for the management team, and reducing reporting time by 50%
- Collaborated with cross-functional engineering and product teams to deliver 10+ end-to-end data science projects, ensuring alignment with business objectives
- Mentored 2 junior team members on ML best practices, code review, and project delivery, contributing to team capability building and knowledge transfer

Data Analyst

January 2022 - October 2022

World Health Organization

- Re-engineered COVID-19 Vaccine Delivery Partnership (CoVDP) Core Analysis Pipeline (CCAP), consolidating fragmented R and Python codebase from multiple organizations into a unified, modularized system that reduced processing time by 67% (from three minutes to less than 1 minute)
- Automated 70% of manual data inputs by integrating 15+ data sources via API-driven ETL pipelines, enabling weekly reporting cadence (improved from bi-monthly) and processing 50GB+ of vaccine implementation data across 190+ countries
- Delivered actionable COVID-19 vaccine insights to WHO, UNICEF, GAVI, and Ministries of Health in 190+ countries, supporting global vaccine equity efforts and informing policy decisions for public and private stakeholders

- Authored comprehensive SOPs for CCAP workflows, ensuring data quality, reproducibility, and seamless knowledge transfer across international partner organizations

Data Science Trainee

March 2021 - June 2021

iLab Africa

- Delivered actionable insights through exploratory data analysis & statistical modeling, driving data-informed decision-making across ten cross-functional projects
- Implemented ETL pipelines and web scraping solutions to aggregate data from multiple sources while maintaining data governance and privacy standards
- Developed and optimized various machine learning models, improving forecast accuracy and reducing error prediction
- Created dynamic Tableau dashboards with KPI tracking, enabling real-time monitoring for the management team

EDUCATION

Master of Science in Data Science and Business Analytics

August 2019 - October 2020

Asia Pacific University of Technology and Innovation | Kuala Lumpur, Malaysia

- GPA 3.6/4.0
- Capstone project: Analyzing Online Sentiments On A Company's Product: A Case Of Safaricom PLC.

Software Engineering

January 2018 - June 2018

Moringa School | Nairobi, Kenya

Bachelor's Degree in Business Information Technology

January 2012 - December 2015

Jomo Kenyatta University of Agriculture and Technology | Nairobi, Kenya

SKILLS

Programming: Python, R, SAS, SQL

Data Science & Statistical Analysis: Data Mining, Data Wrangling, Feature Engineering, Statistical Analysis, Predictive Modeling, Pandas, Numpy, Scikit-Learn

Machine Learning & Generative AI: Machine Learning, Hyperparameter Tuning, Model Evaluation, Natural Language Processing (NLP) Techniques, Large Language Models (LLMs), Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), QLoRA fine-tuning, HuggingFace Transformers

Agentic AI Engineering: AI Agents Development, Agentic AI Workflows, OpenAI Agents SDK, CrewAI, AutoGen, MCP, LangChain, LangGraph

Data Visualization & BI: Tableau, Power BI, Matplotlib, Seaborn

MLOps & Infrastructure: MLOps, Docker Containerisation, CI/CD, Git & GitHub

Project Management & Collaboration: Asana, Notion

Web Development & API: HTML, CSS, Bootstrap, Tailwind CSS, MUI, Django, FastAPI, API integration, RESTful APIs

Core Competencies: Effective Communication, Stakeholder Management, Team Leadership, Cross-Cultural Collaboration, Research & Analysis, Attention to Detail

LICENSES & CERTIFICATIONS

AI Engineer Agentic Track: The Complete Agent & MCP course - Udemy (Ongoing)

AI Engineering Core Track: LLM Engineering, RAG, QLoRA, Agents - Udemy (2025)

SAS Joint Certificate in Data Science and Business Analytics - SAS Institute (2020)